

Simplifying Rational Expressions

Algebra Worksheet · Grade 8–10

Name: _____

Date: _____

Learning Objectives

- Identify and define rational expressions as polynomials in fraction form
- Simplify rational expressions using the laws of exponents, including subtraction of exponents
- Rewrite expressions with negative exponents as positive exponents using reciprocals

Problems

1. Simplify the rational expression by cancelling common factors.

$$\frac{x^4}{x^2}$$

2. Simplify the rational expression using the laws of exponents.

$$\frac{y^6}{y^3}$$

3. Simplify the rational expression. Write your answer with a positive exponent.

$$\frac{a^3}{a^7}$$

4. Simplify the rational expression. Write your answer with positive exponents only.

$$\frac{m^5}{m^{10}}$$

5. Simplify the rational expression with two variables. Write your answer with positive exponents only.

$$\frac{x^6y^4}{x^2y}$$

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6. Simplify the rational expression with a coefficient and two variables. Write your answer with positive exponents only.

$$\frac{6x^5y^3}{3x^2y^5}$$

7. Simplify the rational expression with a coefficient and two variables. Write your answer with positive exponents only.

$$\frac{8a^4b^6}{4a^7b^2}$$

8. Simplify the rational expression with three variables. Write your answer with positive exponents only.

$$\frac{x^9y^4z^5}{x^3y^8z^2}$$

9. Simplify the rational expression with a coefficient and three variables. Write your answer with positive exponents only.

$$\frac{12p^8q^3r^5}{4p^3q^7r^2}$$

10. Simplify the rational expression with a coefficient and three variables. Write your answer with positive exponents only.

$$\frac{15x^{10}y^5z^8}{9x^{14}y^2z^{11}}$$

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Simplifying Rational Expressions — Answer Key

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Answer Key

1. Answer: x^2

- Subtract the exponent in the denominator from the exponent in the numerator: $4 - 2 = 2$
- Result: x^2

2. Answer: y^3

- Subtract the exponents: $6 - 3 = 3$
- Result: y^3

3. Answer: $1 / a^4$

- Subtract the exponents: $3 - 7 = -4$
- We get a^{-4} ; apply the reciprocal rule to make the exponent positive
- Result: $1 / a^4$

4. Answer: $1 / m^5$

- Subtract the exponents: $5 - 10 = -5$
- a^{-5} means $1 / a^5$; apply the reciprocal rule
- Result: $1 / m^5$

5. Answer: $x^4 * y^3$

- Subtract the exponents for x: $6 - 2 = 4$
- Subtract the exponents for y: $4 - 1 = 3$
- Result: $x^4 * y^3$

6. Answer: $2x^3 / y^2$

- Simplify the coefficients: $6 / 3 = 2$
- Subtract x exponents: $5 - 2 = 3$
- Subtract y exponents: $3 - 5 = -2$; move y to denominator to get positive exponent
- Result: $2x^3 / y^2$

7. Answer: $2b^4 / a^3$

- Simplify the coefficients: $8 / 4 = 2$
- Subtract a exponents: $4 - 7 = -3$; move a to denominator to get positive exponent
- Subtract b exponents: $6 - 2 = 4$
- Result: $2b^4 / a^3$

8. Answer: $x^6 * z^3 / y^4$

- Subtract x exponents: $9 - 3 = 6$
- Subtract y exponents: $4 - 8 = -4$; move y to denominator
- Subtract z exponents: $5 - 2 = 3$

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- Result: $x^6 * z^3 / y^4$
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9. Answer: $3p^5 * r^3 / q^4$

- Simplify the coefficients: $12 / 4 = 3$
 - Subtract p exponents: $8 - 3 = 5$
 - Subtract q exponents: $3 - 7 = -4$; move q to denominator
 - Subtract r exponents: $5 - 2 = 3$
 - Result: $3p^5 * r^3 / q^4$
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10. Answer: $5y^3 / (3x^4 * z^3)$

- Simplify the coefficients: $15 / 9 = 5/3$
 - Subtract x exponents: $10 - 14 = -4$; move x to denominator
 - Subtract y exponents: $5 - 2 = 3$
 - Subtract z exponents: $8 - 11 = -3$; move z to denominator
 - Result: $5y^3 / (3x^4 * z^3)$
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